

JESSE KIM

(408) 480-4938 | jessetk267@gmail.com | github.com/jtkim00000 | linkedin.com/in/jesse-kim-b51633308/

EDUCATION

University of Illinois Urbana-Champaign

B.S. Electrical Engineering

GPA: 4.00 / 4.00

Awards: Best Overall Award - UIUC URSA Research Symposium (Spring 2026)

August 2025 - Current

Expected May 2029

EXPERIENCE

Reserach Assistant - GPU Kernel Engineer

May 2026 - Current

- Developing custom CUDA kernels to parallelize a serial Particle-In-Cell simulation, parallelizing kinetic tracking of millions of particles, eliminating a primary computational bottleneck
- Achieved a 24.1x speedup in particle push and a 5.12x speedup in charge deposition using custom kernels
- Optimizing memory throughput by leveraging Struct-of-Arrays data layout for coalesced memory access, and minimizing latency by mitigating PCIe bus transfers and warp divergence
- Validated numerical accuracy of GPU kernel output against serial simulation benchmarks on ion density profiles

Research Assistant - HPC Software Engineer

January 2026 - May 2026

- Designed and benchmarked various optimized 1D Poisson solvers in C++ for a Particle-In-Cell simulation, evaluating stability, CPU time, and error convergence rates
- Tailored diverse numerical methods to a 1D solver matrix, including a Thomas Algorithm and 4 distinct Preconditioned Conjugate Gradient variants (Jacobi, SSOR, Incomplete Cholesky, 1st-Order Neumann series)
- Achieved a 651x speedup in field-solver execution time using the Thomas Algorithm compared to standard Conjugate Gradient and a 13x speedup through iterative SSOR-preconditioning with multidimensional scalability

Illini Solar Car - Firmware Developer

August 2025 - January 2026

- Revised C++ BMS firmware to support implementation of modular PWM duty cycle frequency for fans in battery cooling system, optimizing temperature and power consumption
- Analyzed relationship between fan rotation speed and battery temperature to optimize PWM duty cycle frequency

PROJECTS

Pipelined RISC-V CPU

github.com/jtkim00000/riscv-cpu

- Built a Pipelined RISC-V CPU in Verilog RTL from scratch, implementing 37-instruction subset of RV32I ISA
- Designed custom CPU datapath, first implementing single-cycle CPU, then extending into a 5-stage pipelined CPU (IF/ID/EX/MEM/WB) with forwarding, pipeline stalling, and write-through for full hazard resolution
- Verified CPU functionality across all instruction types, hazard scenarios, and components using hand-assembled RV32I test programs, GTKWave waveform analysis, and isolated module-level testbenches

Noise Detector PCB

github.com/jtkim00000/noise-detector-pcb

- Designed analog signal chain (amplification → peak detection → timed delay → threshold comparison) using LM358 op-amps, LM311 comparator, and Schmitt Trigger Inverter on custom PCB
- Implemented NMOS-switched peak detector with RC hold circuit to capture audio peak, feeding into 1-second output latch via RC delay network, driving 3-level LED indicator display based on 3-level voltage comparator network with potentiometer-tunable references

SKILLS

Programming & Hardware: Verilog RTL, CUDA, C/C++, Assembly(RISC-V, LC-3), PCB Design, Python

Tools: Vivado, GTKWave, Oscilloscope, KiCAD, Git/Github